Each side label is a SQUARED vector length

Orthogonal triangles in observation space; schematic, not to scale.

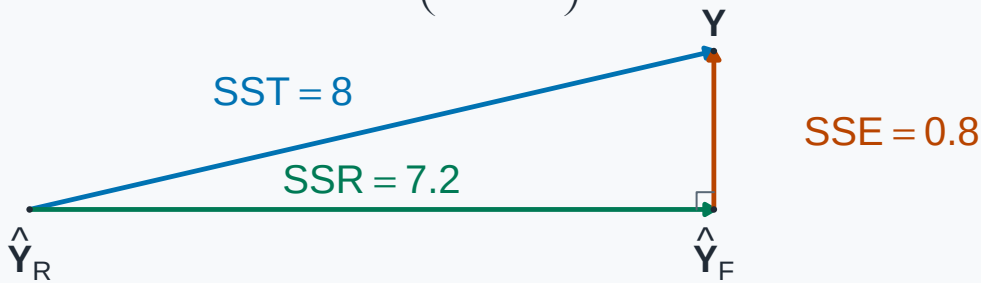
A GENERAL NESTED MODELS

Reduced fitted values can differ across observations.

$$SSE_R = (SSE_R - SSE_F) + SSE_F$$

B SAME FOUR OBSERVATIONS: INTERCEPT ONLY

$$\hat{Y}_R = (3, 3, 3, 3)^T$$



$$8 = 7.2 + 0.8$$